

Nathan Sheng

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EDUCATION

- Cornell University, College of Engineering (class of 2027, major: Mechanical Engineering)
- Corning-Painted Post High School diploma (class of 2023)

EXPERIENCE

- Mechanical Engineer at Mavrik/HavocAI (June 2025 - August 2026, Long Beach, California)
 - Worked with a small team of 7 to develop a heavy-lift drone “Hatchet” and earned a 2 million dollar phase II SIBR government contract to continue to develop it. 6 months after joining we were acquired for 30 million dollars and I worked the next 8 months under HavocAI
- FTC Robotics (9 - 12th grade), team 5484
 - Head builder and driver (11 - 12th grade)
 - Dean’s List Semifinalist (11th grade)
 - Team achievements
 - State Championships Inspire Award (2021 and 2022)
 - World Championships Think Award Finalist (2022)
- Cornell University Autonomous Drone mechanical subteam (sophomore year)
 - Designed and built test stand that measures RPM and force, designed and built camera mount for FPV drone, member of the month for November 2024
- Summer internship at Corning Inc. (2022, 6wks, Corning, NY)
 - Design and development of a demo device for a flexible glass

PROJECTS

- Heavy lift drone “Hatchet” (X8 Drone, 63 in props, 500 lbs weight)
 - Designed avionics mounts and payload systems for Hatchet
 - Highest number of manufacturing, assembly, and maintenance hours on Hatchet
 - Wrote assembly instructions for a mechanical and electrical tech that I had to work closely with to assemble 4x Hatchets
- Toothpick drone (Quadcopter, 3 in props)
 - Entirely designed by me so I could familiarize myself with all parts of making drones
- Autonomous pickup drone (Quadcopter, 10 in props)
 - 2nd drone entirely designed by me; purpose was to be attached to Hatchet at the end of a long-line to guide a mating mechanism onto a cone-shaped target attached to the payload, but it failed
- 2022 FTC Robot
 - My high school magnum opus! Didn’t make it to World’s like the year before, but I had almost complete control over it’s design and I’m proud of it’s macro-layout and it’s cleanness/finish

- Automatic Bird Feeder Camera
 - 10th grade high school project where I made bird feeder with Raspberry Pi to take close up pictures of birds with weight as trigger

SKILLS

- CAD
 - Onshape - Certified Onshape Professional
 - Solidworks - 6 months at HavocAI designing mounts for Hatchet, Corning internship project
 - Fusion 360 - Introduction to Mechanical Design Cornell course
- 3D Printing
 - Assembled Prusa MK3 in 2017, used for robotics and home project parts
 - Managed a print farm for a year that eventually grew to 5x Bambu H2D's and 2x P1S's
- CNC Routing
 - Used Shapeoko 5 Pro CNC to cut at least 10 m² of carbon fiber
- FEA
 - Verified Hatchet landing gear strength
 - Intro course for Ansys Mechanical from SimuTech
- Python
 - Cornell introductory python course, high school courses, high school personal project

REFERENCES

- Fabien Royer, Cornell academic advisor, fr262@cornell.edu
- Eric Gillis, Corning Inc. internship supervisor, gilliser@corning.com
- Patricia Killian, High school robotics coach, killianpa@tasms.com
- Max Owens, former CEO at Mavrik, max.owens@havocai.com